Comparison of the Two Designs

The robot was designed around the following requirements:

- Resistance to wear and repeated handling.

- Watertightness to approximately 10 meters, although initial experiments will take place at depths of only 1–2 meters.

- Easy disassembly for maintenance and component replacement.

- Reproducibility, with minimal adjustment between individual robots.

- A sufficiently low cost to build an entire swarm using the same design.

The main cost differences between the Safe and Low-cost designs concern the mechanical hardware rather than the electronics.

My Recommendation:

The purchasing decision should be made at the scale of the entire swarm. Choosing expensive components for the first robot implies being prepared to purchase the same components for another three or four robots.

I therefore recommend developing and iterating on the Low-cost design during the current project, with particular attention to sealing reliability, durability and reproducibility. This would require more initial development work, but it should make the next stages of the swarm project more affordable and easier to expand.

Details:

Watertight Enclosure

The enclosure is the most significant difference between the two options. Achieving reliable watertightness is one of the main challenges in underwater robot design. The O-ring grooves, mating surfaces and clearances must be accurately dimensioned to ensure consistent seal compression. This requires either precise manufacturing or additional finishing and adjustment.

A ready-made Blue Robotics enclosure largely removes this development work. Its depth rating depends on the selected configuration, but the available ratings substantially exceed our initial operating depth. Another advantage is that Blue Robotics enclosures are widely used in open-source underwater robotics projects, providing useful design references.

Developing our own enclosure with 3D-printed components would require more work to achieve reliable sealing. Possible finishing methods include acetone smoothing for ABS parts and resin coatings, although the exact process remains to be determined and tested.

I expect most of the development effort to be concentrated on the first prototype. Once the geometry and manufacturing process have been validated, they could be reproduced for the remaining robots. Some individual inspection and adjustment would still be necessary, but this approach should significantly reduce the cost of the complete swarm.

Other Differences

Depth sensor. The sensor we already have is not supplied as a waterproof assembly. We would need to waterproof its electronics and integrate it into a sealed, 3D-printed mount while keeping the pressure-sensing interface functional. Epoxy encapsulation is one possible approach. A failure could require replacing both the sensor and its mount, but the overall cost would remain substantially lower than purchasing a ready-made Blue Robotics depth sensor.

Vacuum pump. I do not see a strong reason to purchase the Blue Robotics pump specifically. A lower-cost manual vacuum pump with a suitable gauge and adapter should be sufficient for our leak tests.

Cable penetrators. Standard cable glands appear to be a reasonable low-cost option for our experimental setting, provided their sealing performance is validated at the intended operating depth.

Onboard computer. A Raspberry Pi 3B should be sufficient for the current project because we do not plan to process images onboard. Taka's future projects may require a more recent model, depending on their computational requirements.